

Filip Bělík

filip.belik@utah.edu |  [filipbelik](#) |  [fbelik](#) |  [Google Scholar](#)

PhD Candidate

Department of Mathematics and the Scientific Computing and Imaging Institute
University of Utah, Salt Lake City, UT

EDUCATION

- **University of Utah**

PhD in Mathematics: Applied & Computational Mathematics

August 2022 - Present

Salt Lake City, UT

- Advisors: [Dr. Akil Narayan](#) and [Dr. Christel Hohenegger](#)
- Expected graduation in May 2027
- GPA: 4.00

- **Gustavus Adolphus College**

BA in Mathematics with honors; BA in Computer Science

September 2018 - May 2022

St. Peter, MN

- Cumulative GPA: 3.989
- Major GPA: 4.00

RESEARCH INTERESTS

Numerical analysis, numerical quadrature, model order reduction, differential equations, uncertainty quantification, optimization, and mathematical modeling

TEACHING EXPERIENCE

- **Engineering Calculus II**

Instructor of Record

Spring 2026

University of Utah

- **Engineering Calculus I**

Instructor of Record

Spring 2025

University of Utah

- **Graduate Applied Linear Operators and Spectral Methods**

Teaching Assistant

Fall 2025

University of Utah

- **Graduate Numerical Linear Algebra**

Teaching Assistant

Spring 2024

University of Utah

- **Mathematics of Medicine**

Laboratory Instructor

Spring 2023

University of Utah

- **Mathematics and Computer Science Courses**

Tutor, Teaching Assistant, and Grader

Spring 2019 - Spring 2022

Gustavus Adolphus College

PEER-REVIEWED PUBLICATIONS AND PREPRINTS

1. Henry Crandall, Tyler Schuessler, Filip Bělík, et al. “Cuffless hemodynamic monitoring with physics-informed machine learning models”. In: *Nature Communications* 17.1, 6429 (May 14, 2026). DOI: [10.1038/s41467-026-72693-1](https://doi.org/10.1038/s41467-026-72693-1). URL: <https://doi.org/10.1038/s41467-026-72693-1>
2. Filip Bělík, Akil Narayan, and John Turnage. *Constructive Tchakaloff results and well-conditioned quadrature through randomized least squares*. 2026. arXiv: [2609.09506](https://arxiv.org/abs/2609.09506) [math.NA]. URL: <https://arxiv.org/abs/2609.09506>
3. Filip Bělík, Yanlai Chen, and Akil Narayan. *Greedy Rational Approximation for Frequency-Domain Model Reduction of Parametric LTI Systems*. 2025. arXiv: [2512.23814](https://arxiv.org/abs/2512.23814) [math.NA]. URL: <https://arxiv.org/abs/2512.23814>
4. Filip Bělík, Jesse Chan, and Akil Narayan. *Efficient and Robust Carathéodory-Steinitz Pruning of Positive Discrete Measures*. 2025. arXiv: [2510.14916](https://arxiv.org/abs/2510.14916) [math.NA]. URL: <https://arxiv.org/abs/2510.14916>

ACADEMIC SERVICE AND LEADERSHIP

- **President, University of Utah SIAM Student Chapter** Fall 2024 - Spring 2026
University of Utah
 - Led chapter programming for undergraduate and graduate students, including mathematical modeling competitions, professional development events, and social activities
 - Organized the 2026 [Wasatch SIAM Student Chapters Conference](#)
- **Coordinator, Graduate Peer Mentorship Program** Fall 2025
University of Utah
 - Matched first-year graduate students with current students according to personal and research interests
 - Encouraged regular check-ins addressing academic expectations, qualifying examinations, relocation, and graduate-student well-being
- **Chair, Graduate Student Recruitment Committee** Spring 2025
University of Utah
 - Led a five-member graduate student committee coordinating communication and a campus visit weekend for 23 prospective PhD students
 - Paired prospective students with current graduate students and developed individualized schedules for faculty meetings, academic events, and informal community activities

RESEARCH SOFTWARE

- **[CaratheodoryPruning.jl](#)** Julia
Creator and maintainer
 - Stable, scalable Carathéodory–Steinitz pruning algorithms for constructing positive quadrature rules
- **[ModelOrderReductionToolkit.jl](#)** Julia
Creator and maintainer
 - Projection-based model order reduction methods for parametric dynamical systems

WORK EXPERIENCE

- **University of Utah Mathematics Department** August 2022 - Present
Graduate Research and Teaching Assistant
Salt Lake City, UT, USA
- **Allianz Life** Summers 2021 and 2026
Hedging Intern
Golden Valley, MN, USA
 - Software development with C# and SQL
 - Implemented procedure for automating the labeling of incoming market data
 - Extended existing hedge reporting software
- **Gustavus Mathematics and Computer Science Department** February 2019 - May 2022
TA & Tutor & Grader
St. Peter, MN, USA

HONORS AND AWARDS

- **Outstanding Graduate Student** May 2025
University of Utah Mathematics Department
- **RTG: Optimization and Inversion Summer Grant** May 2023
University of Utah Mathematics Department 
- **Fulbright Canada Mitacs Globalink Research Internship** April 2021
Mitacs Globalink 
- **Hilding Research Fund** April 2020
Gustavus Adolphus College
- **First-Year Research Experience (FYRE)** April 2019
Gustavus Adolphus College 

RESEARCH PRESENTATIONS

- **SIAM Mathematics of Data Science** November 2026
Salt Lake City, UT 
 - Talk: Greedy Frequency Domain Model Reduction for LTI Systems
- **SIAM Imaging Science** November 2026
Salt Lake City, UT 
 - Talk: Hemodynamic Monitoring with a Smartwatch
- **Wasatch SIAM Student Chapters Conference** April 2026
Salt Lake City, UT 
 - Talk: Frequency Domain Greedy Model Order Reduction for LTI Systems
- **Joint Mathematics Meetings** January 2026
Washington, D.C. 
 - Talk: Greedy Rational Approximation for Model Reduction of LTI Systems
- **Wasatch SIAM Student Chapters Conference** April 2025
Logan, UT 
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **MAA North Central Section Meeting** March 2025
Northfield, MN 
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **Model Reduction and Surrogate Modeling** September 2024
La Jolla, CA 
 - Talk: Greedy Frequency Domain Model Reduction for Parametric Systems: New Theory and Algorithms
- **NSF Computational Mathematics PI Meeting** July 2024
Seattle, WA 
 - Poster: Dynamic Bulk Conductivity in Radial Artery
- **Mathematical Opportunities in Digital Twins** December 2023
Arlington, VA 
 - Poster: Dynamic Bulk Conductivity in Radial Artery
- **Midstates Consortium for Math and Science** November 2019
Chicago, IL 
 - Talk: The Propagation of Health-Related Habits on Twitter

TECHNICAL SKILLS

- **Primary Programming Languages:** Julia, Python, C#
- **Additional Programming Languages:** MATLAB, C, C++, Java, SQL
- **Open-Source Contributions:** [UncertaintyQuantification.jl](#), [NonNegLeastSquares.jl](#)
- **Mathematical/Technical Tools:** Jupyter, LaTeX, Maple